

## **Reproducible Processing of Synthetic Study Metadata**

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HSE 711: Foundations in Data Science

[Instructor Name]

[Month Day, Year]

## **Reproducible Processing of Synthetic Study Metadata**

This example shows how a student can organize a short manuscript around a reproducible data-processing exercise. It follows the student-paper setup described by American Psychological Association (2025). Replace the instructional prose and bracketed fields with your own account of the work. The example uses synthetic classroom metadata; it contains no observed research findings.

[Introduce the problem, explain why it matters, and summarize relevant prior work with citations. End this opening section with the research question or objective. For the Week 4 example, the objective is to inspect a metadata file, select records using a documented rule, and verify the resulting files.]

### **Method**

#### **Design and Data Source**

The worked example is a descriptive file-processing exercise using `pseudo_metadata.csv`. The classroom schema has six fields: an unnamed first field, sample identifier, condition, age, sex, and batch. The sex field uses F and M. These labels are source-file encodings and are not inferred from participant names or other variables.

[Report the data source, acquisition date, inclusion criteria, sample size, and relevant permissions. For an empirical study, describe the sampling design and ethics approval or exemption as applicable. For this synthetic exercise, state that the records are simulated and identify the course source.]

#### **Measures and Variables**

[Define the variables relevant to the question, their units, category labels, missing-value conventions, and any derived measures. Distinguish identifiers from measurements. State the source of definitions rather than inventing units.]

#### **Procedure**

The procedure inspects the file header, counts lines, retains the header while selecting F-coded records from field 5, and saves the subset as `only_female.txt`. A copied alias,

`females_metadata.csv`, preserves the second filename used in the exercise. The lab also extracts condition and age into a temporary directory and removes that directory after inspection. The original input is retained.

[Describe the exact software versions, commands or functions, file locations, validation checks, and deviations from the planned procedure. Simple AWK and CUT processing is suitable only for the documented simple CSV layout; a CSV-aware parser is needed for embedded commas or multiline fields.]

### Analysis Plan

[State the descriptive summaries or statistical models used, the treatment of missing data, assumption checks, significance threshold if applicable, and multiple-testing adjustment if relevant. Explain the design rather than listing software packages. The Bash example verifies records and files; it does not require a hypothesis test.]

### Results

[Report only output that was actually produced and checked. Begin with the number of source and eligible records, then present the findings in the order of the research question. Refer to each table and figure in the text. Do not repeat every table entry in prose.]

Table 1 illustrates a results-table layout. Its cells are placeholders, not measured values. Figure 1 illustrates the position of a figure number, title, image, and note. Replace the image area with a verified plot before submission, or remove the figure if unnecessary.

**Table 1**

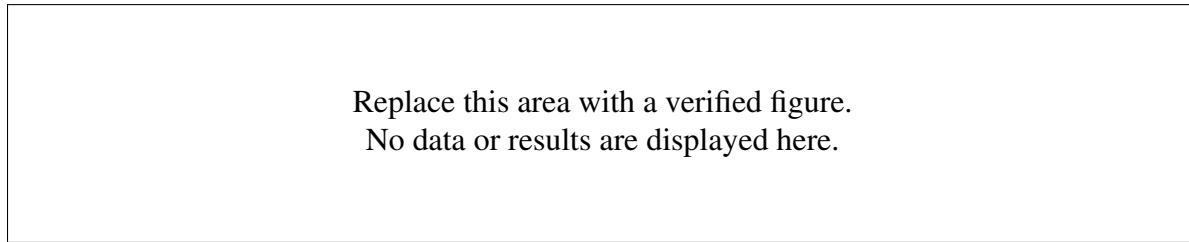
*Metadata Processing and Validation Summary*

| Validation item                     | Observed result         |
|-------------------------------------|-------------------------|
| Source records, excluding header    | [Insert verified count] |
| Selected records, excluding header  | [Insert verified count] |
| Header retained                     | [Insert check outcome]  |
| Condition and age extracted         | [Insert check outcome]  |
| Subset files retained after cleanup | [Insert check outcome]  |

*Note.* Record counts exclude the header; a newline count may include it. Replace every cell after inspecting the actual outputs. The table describes processing checks, not population estimates.

## Figure 1

*Example Figure Area for a Verified Analysis Plot*



*Note.* Describe the observations, axes, units, groups, scale, and abbreviations needed to read the figure. State whether the data are simulated. Cite an adapted or reproduced figure and check its permission terms.

## Discussion

### Principal Findings

[Answer the research question using the verified results. Explain the meaning of the findings, compare them with prior work using appropriate citations, and distinguish a descriptive pattern from inferential evidence.]

### Limitations

[Explain limitations of the design, input quality, missing data, assumptions, and generalizability. For this example, synthetic records cannot establish population findings, and simple delimiter tools do not implement general CSV parsing. Do not frame successful file creation as proof of scientific validity.]

### Implications and Conclusion

[Explain the practical or methodological implication and the next justified step. Close with a concise answer to the objective; avoid introducing new results or making claims beyond the study design.]

The student title-page guide provides an additional formatting reference (American Psychological Association, n.d.). In your manuscript, move substantive citation discussion into the relevant text section; this sentence is included to demonstrate a parenthetical citation. Delete it when replacing these instructional sources.

### References

American Psychological Association. (n.d.). *Title page setup*.

<https://apastyle.apa.org/style-grammar-guidelines/paper-format/title-page>

American Psychological Association. (2025). *Student paper setup guide*.

<https://apastyle.apa.org/instructional-aids/student-paper-setup-guide.pdf>

## Appendix

### Reproducible Command Example

This example is run from the repository root with the course file in the ignored data/ directory. The fixture test creates its own synthetic input. The lab's temporary directory is disposable; verify that the subset outputs remain after the lab finishes.

```
# Validate the lab using its self-contained fixture.
bash "03_Group_Work/Week_4_Bash_Lab/test_week4.sh"

# Run after placing the course CSV in data/.
bash "03_Group_Work/Week_4_Bash_Lab/run_week4.sh" \
  data/pseudo_metadata.csv ./week4_practice

# Inspect retained output; line counts include a header.
head -n 5 week4_practice/only_female.txt
wc -l week4_practice/only_female.txt
```

[Record the operating system, Bash version, source-file checksum, execution date, and actual validation outcomes. Remove this appendix if supplementary commands are not needed for the assignment.]